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Creating Electronic Books with Code and Artificial Intelligence Assistants

Course for the Faculties of Sciences and Chemical Sciences

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Welcome to **Creating Electronic Books with Code and Artificial Intelligence Assistants**.

What is this?

This is a sample workspace illustrating the pedagogical power of interactive TeachBooks / Jupyter Books designed for the teaching staff of the **Faculties of Sciences and Chemical Sciences at USAL**.

In this simplified configuration, we have enabled a single illustrative chapter to demonstrate how to structure advanced scientific explanations by combining rich text, complex LaTeX mathematical formulas, and interactive MyST directives.

Available Content

- *The Central Limit Theorem*: A practical example of interactive scientific writing.

PDF Version

You can also download the printable, formatted version of the book:

- [Download PDF in English](#)
- [Descargar PDF en español](#)

Note

This interactive project is optimized for web viewing and PDF compilation, preserving the highest typographical quality using LaTeX.

Part I

Sample Chapter

THE CENTRAL LIMIT THEOREM

The **Central Limit Theorem (CLT)** is one of the most fundamental and remarkable results in probability theory and statistics. This theorem explains why the normal distribution (or Gaussian bell curve) appears so frequently in nature, science, and engineering.

In this chapter, we will explore its formal definition, its practical implications, and observe how it behaves as we increase the sample size.

1.1 Formal Definition

i Central Limit Theorem

Let $X_1, X_2, \dots, X_n$ be independent and identically distributed (i.i.d.) random variables with finite population mean μ and finite standard deviation $\sigma > 0$.

If we define the sample mean as:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Then, as the sample size n approaches infinity ($n \rightarrow \infty$), the distribution of the standardized variable Z_n converges in distribution to a standard normal distribution $\mathcal{N}(0, 1)$:

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1) \quad (1.1)$$

As shown in equation (1.1), the distribution of the sample mean approaches a normal distribution with mean μ and variance σ^2/n , regardless of the original probability distribution of the individual random variables X_i .

1.2 Pedagogical Implications

Why is this theorem so important in teaching and scientific practice?

1. **Statistical Inference:** It allows approximating and inferring population parameters (such as means and proportions) using the normal distribution, even if the original population is not normally distributed.
 2. **Confidence Intervals:** It provides the theoretical foundation for calculating confidence intervals and performing traditional hypothesis tests for sufficiently large samples (usually $n \geq 30$).
 3. **Simplicity:** It simplifies complex mathematical models by approximating the sum of many independent random effects with a single Gaussian distribution.
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1.3 How does it behave in practice?

Reflection Question: How large must the sample size n be?

The rate of convergence toward the normal distribution depends directly on the symmetry of the original distribution of the random variables:

- **Symmetric Distributions** (such as the uniform distribution): Converge extremely fast. Even with small samples like $n = 5$ or $n = 10$, the sample mean already exhibits a shape very similar to the normal curve.
- **Asymmetric or Skewed Distributions** (such as the exponential or Pareto distribution): Require significantly larger sample sizes ($n \geq 30$ or even $n \geq 100$) for the original skewness to disappear from the sample mean.

This distinction is crucial when designing experiments and statistical analyses in science classrooms.

1.4 Conceptual Visualization of the Theorem

Let's imagine rolling a standard 6-sided die. The probability distribution of getting any score from 1 to 6 is uniform (all outcomes have a probability of $1/6$).

- **If we roll 1 die** ($n = 1$): The distribution is completely flat (uniform).
- **If we roll 2 dice and average their values** ($n = 2$): The distribution of the average has a triangular shape (3.5 is the most probable outcome).
- **If we roll 10 dice and average them** ($n = 10$): The resulting distribution is a smooth bell curve, almost indistinguishable from a perfect normal distribution.

Part II

References

BIBLIOGRAPHY